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Capturing Human Perception of Level Difficulty in a Spatial Bridge Building Game using a Qualitative Solver

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Abstract

This paper presents an analysis of the factors that affect human perception of level difficulty in the Camelot Jr. bridge building puzzle game. Camelot Jr. challenges players across spatial reasoning tasks focused on physical support and structural stability. The goal of the game is to create a continuous walkable path between two avatars using a limited inventory of building blocks. As these avatars are situated at different vertical elevations and the available blocks often cannot span the empty space directly, players must discover different object affordances within the environment to construct stable, multi-level bridges. We develop a hybrid qualitative solver for the game to computationally examine its complexity. The solver leverages center-of-mass stability logic to analyze how to proceed to the next step in the solution. With an inherent symbolic grounding, the solver computes various dynamic factors that affect challenge difficulty. Using this solving technique, we identify a range of features spanning initial and final grid configuration and solution complexities that affect human perception of this task's challenge. We validate these metrics using the difficulty ranking of 48 expert-labeled challenges from the Camelot Jr. developers. We also conduct a user study to triangulate and present preliminary results comparing these metrics against human-perceived level difficulty. This study revealed that physical spatial properties—specifically, the vertical elevation of the two avatars ($r = 0.657, p = 0.039$))—show a statistically significant positive correlation with human-perceived difficulty, alongside strong positive trends in count of vacant grid cells ($r = 0.538, p = 0.109$) and log-transformed stable configuration complexity ($r = 0.569, p = 0.086$). To the best of our knowledge, this is the first work to identify and validate level-difficulty metrics that analyze human perception of task difficulty in bridge-building puzzle games.

1. Introduction

As games grow in user base, there is increasing interest in using them to understand aspects of human behaviour (Isbister & Schaffer, 2008; Thompson et al., 2013). They offer a controlled environment with high complexity and can create situations similar to real life that challenge players.

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They have served as a test-bench to develop algorithms((Mnih et al., 2015; Silver et al., 2016; Carroll et al., 2019) and the gameplay data has been used to analyze human parameters (Coutrot et al., 2018; Thompson et al., 2014) and train human skills (Merriman et al., 2022; Kimura & Moussavi, 2021; Giakis et al., 2018; Milani et al., 2019). Games provide massive data-collection opportunities, can create personal learning opportunities in an immersive and engaging way (Nacke et al., 2025; Plass et al., 2015), and provide an accessible way to collect data in both physically observed and unobserved distant settings.

One class of games where large-scale human data is collected is puzzle games (Pelánek, 2014; Coutrot et al., 2019a; Falomir et al., 2021). They have been deployed using web and mobile phones to reach to a much larger audience. Spatial reasoning games are a prominent category in this field, currently attracting significant attention for their educational advantages. Spatial cognition studies have shown that students with better spatial skills are more successful in STEM disciplines (Newcombe, 2010). Spatial reasoning is not an innate ability, since it can be trained (Sorby, 2009) and showed a lasting performance (Uttal et al., 2013). Action video games modify visual selective attention Green & Bavelier (2003), and promote better processing speed and reaction time Owsley et al. (2018). Such educational games allow players to actively engage in spatial reasoning, strengthening their ability to understand complex spatial relationships and transformations Garcia-Segarra et al. (2024). Moreover, developing these games with human-like reasoning elements can make their workings more explainable Cardona-Rivera & Young (2013); Jaiswal & Srivastava (2024); Lake et al. (2017).

However, it is important to make these games engaging and immersive. For this, challenging players with levels of appropriate difficulty is of utmost importance (Schmierbach et al., 2014; Kiili et al., 2014; Hendrix et al., 2018). Therefore, problem generation in puzzle games is an important field of study (Jarušek & Pelánek, 2011b; Linehan et al., 2014; Kotovsky et al., 1985; Spierewka et al., 2021) and it becomes necessary to understand what constitutes a difficult puzzle.

We explore this problem through the lens of Camelot Jr., a spatial puzzle game designed to train human spatial skills in stability perception and spatial planning. Specifically, we focus on how to evaluate parameters that make the puzzle levels difficult. For this, we develop a solver model grounded in human-like commonsense logic that combines qualitative reasoning modules with physical stability calculations to compute various computational heuristics for solving time. In addition, we identify many initial and solution-grid metrics and conduct a web-based study using expert labels, human ratings of level difficulty, and response time to validate them as measures of human perception of puzzle difficulty.

The rest of the paper is organized as follows. Section 2 presents related work on using games in teaching systems and evaluating their level of difficulty. Section 3 outlines the Camelot Jr. videogame. Section 4 explains the various elements of our puzzle solver. Section 5 analyses our results on level difficulty assessment. Section 6 concludes with a discussion on current and future work.

2. Related Works

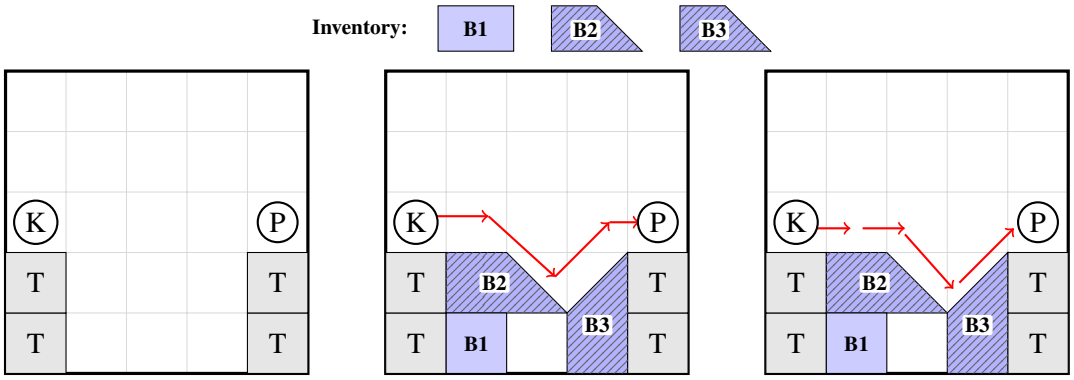
The literature shows that game-based approaches have been effective as tools for improving spatial skills (Guo et al., 2023; Merriman et al., 2022; Coutrot et al., 2019b; Kimura & Moussavi, 2021; Falomir et al., 2021; Garcia-Segarra et al., 2024; Falomir & Oliver, 2016). For visuospatial skill training in children, games benefit the training of visualization, perception, and rotation skills, to name a few (Kuznetcova et al., 2023; Pekarkova & Milkova, 2017; Lin & Chen, 2016). Training these skills early can improve spatial functioning and reduce socio-economic status disparities in a technology-driven society (Newcombe & Frick, 2010). Moreover, if adults train spatial skills, they can also improve attention (Zelinski & Reyes, 2009).

However, spatial reasoning regarding structural stability remains unexplored in human skill-training games. Visual stability perception narrows the gap between abstract visual geometry and the physics of the real world. It is necessary to navigate the physical risks around us and develop our own intuitive physics model. Camelot Jr. provides a unique combination in this domain, integrating physical stability reasoning with knowledge of affordances associated with various game objects.

To make problem-solving more interesting, estimating problem difficulty is crucial. The difficulty of an algorithmic puzzle is related to the player's enjoyment and thus affects the user experience. Thus, puzzle difficulty is directly related to human performance, and ordering challenges from easy to difficult becomes an important task in any learning system (Scheiter & Gerjets, 2019). To make progress on this task, web-based platforms are an opportune environment for collecting a large-scale dataset. For the Sudoku puzzle, Pelánek (2014) collected a massive dataset from multiple web server gameplays to analyze human perceptions of difficulty in Sudoku puzzles. Other common games analyzed for the same task include puzzle games such as Tower of Hanoi (Kotovsky et al., 1985), Sokoban (Jarušek & Pelánek, 2011a), and Rush hour (Jarušek & Pelánek, 2011b), to name a few. Using a different route, many methods have used an algorithmic solver to estimate the difficulty of logic-grid puzzles (Shyne et al., 2024; Valenzuela et al., 2025; Olugbade et al., 2022). In this work, we combine heuristic metrics with an automated qualitative solver and test the resulting approach using a web-based user study.

In the user study for human perception of puzzle difficulty, two approaches are observed. Many studies have used user rating as a means to label puzzle difficulty (Van Kreveld et al., 2015; Chen et al., 2023), and others have used response time to measure problem difficulty (Pelánek, 2017; Effenberger et al., 2019). Problem-solving time is an interesting indirect approach and has been heavily explored in the intelligent tutoring systems domain Jarušek & Pelánek (2012).

To develop puzzles that are interesting to humans, bringing human-like thinking into puzzle generation is important, as computer-generated problems often lack it (Browne, 2015). Thus, we hypothesize that capturing human perceptions of difficulty could make procedural generation more human-appealing. Lynch (2021) have tried to integrate such human-like solving strategies in Sudoku. Through our qualitative solver and metrics-based approach, we also try to follow the same path.



(a) Initial fixed configuration.

(b) Knight's solution path.

(c) Agent's internal graph traversal.

Figure 1: Qualitative discrete representation of a *Camelot Jr.* game level layout with one rectangular block B1 and two stair blocks B2 and B3: (a) initial problem state, (b) a stable solution path with stair-blocks placed in horizontal and vertical orientations, and (c) the agent's internal path traversal graph. The agent's moves are: (*move-right*), (*move-right* + *move-diagonal-down*), and (*move-diagonal-up*).

3. The Camelot Jr. Game: Background

All gameplay in *Camelot Jr.* occurs on a vertical 2D *Grid* (G) that represents the castle layout, where the *Ground* serves as the base of the grid and acts as the primary immutable support for all structures. The core task involves two animated characters, the *Knight* (K) and the *Princess* (P), who must be connected by a path composed of various block pieces across the layout. Figure 1 shows a schematic representation of a typical level in Camelot Jr.. The environment contains fixed *Towers* (T), which are solid, permanent structures positioned at the start of a puzzle. To bridge structural gaps, the player deploys pieces from a limited inventory of *Blocks* (B), consisting of rectangular building blocks that serve as stable, walkable platforms and stair-blocks, which combine a solid block with a ramp side that allows diagonal movement up or down. The ramp side occupies exactly one grid cell. The rectangular blocks can be of sizes 1 to 3. The stair-block has one ramp side combined with rectangular blocks of size one or two.

The entire environment is governed by a global *Gravity* constraint, ensuring that every placed block structure is physically viable under static equilibrium to be considered a valid solution. Together, these elements dictate the available *Walkable Space*, defined as the set of empty grid cells located immediately above stable structures or the ground. Crucially, for stair-blocks, the ramp face itself constitutes the active walkable surface rather than the empty grid cell above it.

Levels vary in placement of the *Knight* and the *Princess*, presence of *Towers* at different locations in the grid space, and an inventory of *Blocks* to build the bridge. To solve the level, all of the inventory items must be placed in the game.

Table 1: Qualitative orientation labels and the supported affordances for inventory blocks. * for $\chi = 1$.

Label	State Name	(Axis, f, ι)$*$	Functional Affordance
HN	Horizontal Normal	$(H, 0, 0)$	Flat structural support + Downward ramp from left to right
HM	Horizontal Mirror	$(H, 1, 0)$	Flat structural support + Upward ramp from left to right
HNI	Horizontal Inverted	$(H, 0, 1)$	Flat structural support
HMI	Horizontal Mirror Inverted	$(H, 1, 1)$	Flat structural support
VN	Vertical Normal	$(V, 0, 0)$	Upward ramp from left to right
VM	Vertical Mirror	$(V, 1, 0)$	Downward ramp from left to right
VNI	Vertical Normal Inverted	$(V, 0, 1)$	Inverted Pillar (Illegal placement)
VMI	Vertical Mirror Inverted	$(V, 1, 1)$	Inverted Pillar (Illegal placement)
RB	Rectangular Solid Block	$\chi = 0$	Flat structural support

The game follows real-world physics, with each top-level element requiring structural support from below and stability under gravity. In the end, the level is solved if the players build a stable path from the *Knight* to the *Princess* using the provided inventory.

4. Symbolic Reasoning Solver

We follow an explainable, human-like spatial solving procedure while developing an automatic level solver. Grounded in symbolic modules, the system can compute various metrics, such as the total number of stable arrangements of inventory blocks and the number of ways to place them in the grid. As we show later, such metrics affect human perception of level difficulty in bridge-building puzzles like Camelot Jr.

4.1 Grid Environment (G)

The grid is a set of cells $c_{x,y}$ where $x \in [0, W)$ and $y \in [0, H)$, with W and H being the width and height of the grid:

$$c_{x,y} = \begin{cases} 0 & \text{Empty Space} \\ 1 & \text{Tower } (T) \\ id \geq 1 & \text{Placed Block } (B) \end{cases}$$

$K = (x_k, y_k)$ (Knight) and $P = (x_p, y_p)$ (Princess).

4.2 Block Definition

A block B is defined as a 4-tuple representing its intrinsic properties and current state:

$$B = \{N, \text{id}, \mathcal{O}, \chi\}$$

Where:

- $N \in \mathbb{Z}^+$ is the **Scalar Size**, representing the length of the block segment.

- id $\in \mathbb{Z}$ is the unique identifier used for tracking grid occupancy and enforcing placement order.
- $\mathcal{O}$ is the **Orientation State Vector**, stores the particular orientation of the block to be used for grid placement.
- $\chi \in \{0, 1\}$ is the **Stair Indicator**, defining whether the block possesses ramp side ($\chi = 1$) or is a standard flat block ($\chi = 0$).

A placement is a mapping $L : B \rightarrow (x, y)$, where (x, y) represents the anchor coordinate (the bottom-left cell) of the block in the grid $\mathcal{G}$. We define $B_{x,y}$ as the block instance whose anchor point is situated at (x, y).

4.3 The Orientation State Vector

The vector $\mathcal{O} = (\text{Axis}, f, \iota)$ governs the spatial projection and parity of the block via three qualitative operations:

- **Axis** $\in \{H, V\}$: Determines if the block is Horizontal or Vertical.
- **Reflection** ($f \in \{0, 1\}$): Governs horizontal parity (Mirroring).
- **Inversion** ($\iota \in \{0, 1\}$): Governs vertical parity (Flipping).

The solver navigates the orientation space using three discrete operations:

- **Reflect (R):** A Horizontal Flip (Reflection across the vertical axis),
- **Invert (I):** A Vertical Flip (Reflection across the horizontal axis),
- **Transpose (T):** A 90° Anti-Clockwise Rotation.

Table 1 describes the various orientations for a stair-block, along with the affordance of the orientations. Figure 2 shows a schematic of the various ramp orientations. Horizontal Normal refers to the base orientation of a stair-block with the ramp occupying the rightmost size (1st diagram in Figure 2).

For a rectangular block denoted as **RB**, only two distinct orientations exist - Horizontal and Vertical and we use only the Axis label to define them. Using these block manipulations, we can compute the various transformations they can undergo during the solution phase, which allows us to explicate the computational complexity of the solutions.

4.4 State Transition Matrix

Table 2 defines how orientation changes through various qualitative operations for a stair-block. The solver utilizes this to track valid placements while searching for solutions. For rectangular blocks, only to-and-fro Horizontal-Vertical state transitions exist. For subsequent discussions, we consider only the legal placements of stair-blocks.

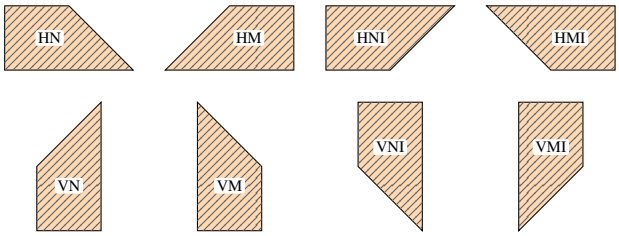


Figure 2: The eight qualitative orientation states for a stair-block ($N >= 2$). Arrangements like VNI and VMI are illegal as per the game rules.

Table 2: State Transition Matrix for qualitative block orientations. Pruned states are marked as Illegal.

Initial State	**Reflect** (R)	**Invert** (I)	**Transpose** (T)
HN	HM	HNI	VN
HM	HN	HMI	VNI (Illegal)
HNI	HMI	HN	VM
HMI	HNI	HM	VMI (Illegal)
VN	VM	VNI (Illegal)	HMI
VM	VN	VMI (Illegal)	HM

4.5 Qualitative Spatial Reasoning Solver

To ensure we capture human-centric game definitions, we use a discretized representation of the game world into a partially filled 2D grid with usable inventory blocks, as previously described. The agent observes the current state of the game, identifies possible placements of an inventory item and then tests these placements to check the *Knight* and *Princess* reachability. A human player would also follow a similar strategy where they assess the current state of the grid and inventory, analyze the next valid placement options for inventory items and test out their assumptions, often backtracking their path.

The agent's strategy is implemented using a recursive protocol. Similar to a player who would rule out any placement unlikely to reach the goal cell (P), our solver implements horizontal and vertical pruning strategies to cut off impractical paths that do not reach the goal with the remaining inventory. Specifically, the solver imposes spatial constraints to enforce placements within the grid. Similarly, to optimize the search process, the solver implements support constraints by evaluating a block's immediate support before committing to a placement. Following this approach recursively, the agent identifies all valid inventory placements that connect the *Knight* to the *Princess*.

Moreover, following this strategy, the solver computes properties associated with various game elements, such as their possible positions, orientations, transformations, and how they affect the branching complexity for a human solver. We use these heuristics in our user study to analyze how a level with varying complexities across these properties poses a challenge for a human player.

Centre-of-Mass (COM) based Stability Analysis: The solver integrates quantitative COM arithmetic to verify structural stability for the whole system of placed blocks. It follows a top-down recursive analysis on load propagation to ensure that each placement of a block in the final solution remains stable. This calculation should be checked only after placing all inventory blocks, as an initial unstable arrangement of blocks can sometimes be stabilized by subsequent counterweight placements. To enable this facility, the solver calculates the weights and moments of each substack of blocks, beginning from the top, and then it checks whether the resultant COM lies within the boundaries of its supporting blocks.

5. Factors Affecting Human Perception of Level Difficulty

Prior research has classified puzzles into three broad categories based on *Initial* and *Solution* grid configurations and *Dynamic features* for difficulty assessment metrics (Van Kreveld et al., 2015; Spierewka et al., 2021). Following a similar rationale, we identify various metrics that can influence the complexity of a path-building puzzle, such as Camelot Jr., spanning these three categories (Table 3). These metrics cover dimensions of human cognitive load and computational search space, emphasizing the categories of initial and solution grid configurations, as well as the factors influencing the solving phase. Other puzzles, following a similar framework, may benefit from our specific heuristics.

We acknowledge that there may still be additional factors relevant to difficulty assessment in such games. Nonetheless, this work takes an initial step to validate the efficacy of these metrics. For this purpose, we examine these metrics using multiple strategies to better understand their relation to human perception of puzzle difficulty.

5.1 Level Difficulty Analysis Using Expert Labels

Game developers for Camelot Jr. provide a list of 48 levels of increasing difficulty to challenge the players. We use these expert-labeled levels as ordinal ranks to assess our metrics. While some levels may not always present a monotonic increase in difficulty, the trend is clear in the problems: they systematically train players from easier to more difficult puzzles. Since the difficulty rankings are ordinal, we examine how level difficulty compares with various metrics using Spearman's coefficients and also present results using Pearson's coefficients.

Table 3 maps the broad factor categories to functional metrics and details their Spearman correlation across the 48 levels in Camelot Jr. videogame. This strategy shows that all the metrics tested have a significant correlation with the difficulty rankings. Specifically, dynamic factors such as the *Stable Configs.* ($\rho = 0.78$), which capture the various choices of stable inventory arrangement present during the solution search, show a stronger correlation with the expert-labeled ranks. Following closely are initial grid factors, such as *Inventory Size* ($\rho = 0.72$) and the *Total Volume* ($\rho = 0.69$) of empty grid cells, which directly affect block placement options throughout the level. Other initial factors, such as *Initial Placement Entropy (IPE)* ($\rho = 0.63$), which capture how blocks are placed in the starting grid, indirectly measure the breadth of placement decisions based on the initial setup.

Table 3: Various metrics tested for human perception of level difficulty. Correlation mapping ($[r, \rho]$ represents Pearson r and Spearman ρ) over the 48 expert labeled levels. ($^{***}p < 0.001$, $^{**}p < 0.01$).

Factor Category	Quantitative Metric	Description
Initial Factors	Inventory Size $[r = 0.70^{***}, \rho = 0.72^{***}]$	Measures the raw count of available inventory pieces.
	Total Volume $[r = 0.68^{***}, \rho = 0.69^{***}]$	The county of empty grid cells in the level.
	Height Sum $[r = 0.61^{***}, \rho = 0.59^{***}]$	The sum of the start and goal vertical coordinates ($K_y + P_y$).
	Large Pieces $[r = 0.45^{**}, \rho = 0.45^{**}]$	Total number of multi-cell wide/tall blocks (e.g., 3×1).
	Initial Placement Entropy (IPE) $[r = 0.64^{***}, \rho = 0.63^{***}]$	Total number of valid ways of legally placing and inventory piece on the initial grid
	Unique Shapes $[r = 0.61^{***}, \rho = 0.61^{***}]$	The count of distinctly shaped blocks in the inventory affects the complexity of the combined affordance provided by the blocks.
Dynamic Factors	Log(Stable Configs. + 1) $[r = 0.78^{***}, \rho = 0.78^{***}]$	Total number of stable inventory arrangements possible while trying all arrangements of inventory blocks in the search space.
	Maximum Decision Breadth (MDB) $[r = 0.50^{***}, \rho = 0.58^{***}]$	Measures the maximum of all local branching factor (placement choices) at any instance while solving the level.
Solution Factors	Total Solutions $[r = 0.43^{**}, \rho = 0.51^{***}]$	The number of valid paths to success.

Unique Shapes (ρ = 0.61) captures the various affordances available to the players in terms of different shapes and sizes of the blocks. Similarly, *Height Sum* (ρ = 0.59) of the vertical elevations of the *Knight* and the *Princess* affects the complexity of the problem by requiring the creation of multi-level stable structures for the path. The second dynamic factor we tried, *Maximum Decision Breadth* (ρ = 0.58), evaluates the maximum number of placement choices across all solution moves and directly measures the computational complexity of the level.

However, it is quite possible that some of these features reflect the same latent parameter. Therefore, to avoid redundant features, we analyzed the Spearman inter-correlation of these parameters (Table 4). The results show that *Inventory Size*, *Total Volume*, and *Initial Placement Entropy* exhibit high correlation among themselves, all belonging to the category of initial grid factors. Both the factors in the dynamic factors category — *Stable Configs.* and *Maximum Decision Breadth (MDB)* — show strong correlation (ρ = 0.792). Surprisingly, *Stable Configs.* also shows strong correlation

Table 4: Pairwise feature correlation matrix (Spearman's ρ) for the 48-level dataset. MDB = Maximum Decision Breadth, IPE = Initial Placement Entropy

Feature	Inventory Size	Total Volume	Height Sum	Stable Configs.	MDB	Total Solutions	Large Pieces	Unique Shapes	IPE
Inventory Size	1.000	**0.876**	0.244	**0.811**	0.517	0.672	0.374	0.671	**0.902**
Total Volume	**0.876**	1.000	0.255	0.780	0.575	0.595	0.688	0.616	**0.898**
Height Sum	0.244	0.255	1.000	0.355	0.089	0.135	0.227	0.354	0.093
Stable Configs.	**0.811**	0.780	0.355	1.000	0.792	0.687	0.384	0.556	**0.816**
MDB	0.517	0.575	0.089	0.792	1.000	0.592	0.307	0.293	0.651
Total Solutions	0.672	0.595	0.135	0.687	0.592	1.000	0.035	0.026	0.793
Large Pieces	0.374	0.688	0.227	0.384	0.307	0.035	1.000	0.596	0.360
Unique Shapes	0.671	0.616	0.354	0.556	0.293	0.026	0.596	1.000	0.417
IPE	**0.902**	**0.898**	0.093	**0.816**	0.651	0.793	0.360	0.417	1.000

with *Inventory Size* and *Initial Placement Entropy*, even though it is measured dynamically during the process of solution finding.

Height Sum appears to be fairly uncorrelated with the other features. *Large Pieces* and *Unique Shapes* show some inter-correlation stemming from the fact that an inventory set may not have many large pieces, which means having large pieces also increases the count of unique shapes. Interestingly, they both also correlate with *Total Volume*; however, the first set is due to inventory blocks, whereas *Total Volume* is a parameter of the *Grid* configuration, and *Towers* are present.

5.2 Level Difficulty Analysis Using Player Ratings

To get players to rate each level for difficulty analysis, we ran a web-based test in which players solved 10 problems with different grid and inventory configurations. The players would complete a level and then rate their perceived difficulty on a Likert scale from 1 to 7. We also allowed them to skip a challenge if it was too difficult for them. To avoid spurious data, we removed those ratings where participants skipped a challenge within 8 seconds, suggesting they did not think through the problem (the minimum solution time in our collected data was 8 seconds).

Each player was shown the 10 problems in a random order to avoid a monotonic increase in difficulty. This randomness was introduced to mitigate biases from level sequencing and fatigue carryover, as it can reduce bias in collected data arising from contextual effects (Pelánek et al., 2022). Some players chose not to play all levels, and we received their ratings for some of the levels.

A total of 15 university students participated in the experiment. 5 of them completed all levels. Overall, we got 77 ratings across the 10 levels after data cleaning. For these data points, the mean and standard deviation for difficulty rating were 3.9 ± 2.1.

Table 5 shows the Spearman (ρ) and Pearson (r) correlation coefficients for our proposed metrics against player ratings. With $N = 10$ levels, statistical power is limited—requiring $|r|, |\rho| \geq 0.632$ to reach $p < 0.05$. Even so, several metrics still show strong positive trends ($r, \rho > 0.50$). Among initial grid factors, *Height Sum* ($r = 0.657, p = 0.039$), *Total Volume* ($r = 0.538, p = 0.109$), and

Inventory Size ($r = 0.428, p = 0.217$; $\rho = 0.454, p = 0.187$) show moderate-to-large correlations with human ratings. The dynamic factor *Log(Stable Configs. + 1)* also shows a strong positive correlation ($r = 0.569, p = 0.086$).

Among these factors, *Height Sum* shows the largest correlation and is the only statistically significant result ($p < 0.05$). This is likely because, beyond the inventory specifics, the height of the *Knight* and *Princess* is the most visible parameter for the level. To check this point, we ran an in-game survey with six participants. When asked which factors they considered before building a path, the height of the *Knight* and *Princess* received the most votes, corroborating *Height Sum* as a prominent perceptual signal of difficulty.

Table 5: Correlation Coefficients Between Level Metrics and Human-Rated Difficulty ($N = 10$).

Feature	**Pearson Correlation (r)**		**Spearman Correlation (ρ)**	
	Coefficient	**p-value**	**Coefficient**	**p-value**
Inventory Size	0.428	0.217	0.454	0.187
Total Volume	0.538	0.109	0.488	0.153
Height Sum	**0.657***	**0.039**	0.546	0.103
Log(Stable Configs. + 1)	0.569	0.086	0.297	0.405
MDB	0.230	0.524	0.012	0.973
Total Solutions	-0.084	0.818	-0.148	0.683
Large Pieces	0.287	0.421	0.234	0.516
Unique Shapes	0.371	0.291	0.440	0.203
IPE	0.236	0.512	0.212	0.556

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

5.3 Level Difficulty Analysis Using Time Spent on Each Level

Previously, we compared players' subjective difficulty ratings to evaluate our proposed metrics. Subjective ratings, such as in our case, can suffer from hindsight bias once a players solves a problem. Consequent playing may also cause memory distortion and contrast bias, in which one very difficult problem makes the next feel easier. While randomization mitigates this to some degree, we also conduct a more objective evaluation. To this end, previous research has used response time as a measure of task difficulty (Jarušek & Pelánek, 2011b; Pelánek & Jarušek, 2015).

Following a similar rationale, we compare the total player time spent on each level against our structural difficulty metrics (Table 6). While the sample size ($N = 10$) constrains statistical significance at the $p < 0.05$ threshold, several metrics exhibit large effect sizes with clear positive trends. Specifically, *Log(Stable Configs. + 1)* demonstrates the strongest linear correlation with completion time ($r = 0.619, p = 0.056$), while *Height Sum* displays the strongest rank-order correlation ($\rho = 0.621, p = 0.055$; $r = 0.553, p = 0.098$). Physical scale and shape variety metrics—including *Unique Shapes* ($r = 0.525, p = 0.120$), *Inventory Size* ($r = 0.502, p = 0.140$), and *Total Volume* ($r = 0.478, p = 0.163$)—also show consistent moderate-to-large positive associations. *Large Pieces*, *MDB*, and *IPE* do not show strong correlation values with this amount of data.

The average completion time and standard deviation for a challenge were 82 seconds and ± 136 seconds.

Solving time for difficulty perception can also lead to errors. For example, a player may take a long time at a level only to realize the solution is not so difficult; hence, their subjective rating of difficulty is not very high, but the objective time spent suggests the problem is very difficult. Similarly, if a player perceives a problem to be very difficult and skips it early on, providing a high difficulty rating, still the objective time spent is less causing a deviation from expected trends. As we collect more data, we believe this comparison would show more similarity with our previous results.

Table 6: Correlation Coefficients (Pearson's r and Spearman's ρ) between level metrics and Player time spent on each level ($N = 10$).

Feature	**Pearson Correlation (r)**		**Spearman Correlation (ρ)**	
	Coefficient	**p-value**	**Coefficient**	**p-value**
Inventory Size	0.502	0.140	0.376	0.284
Total Volume	0.478	0.163	0.348	0.324
Height Sum	0.553	0.098	0.621	0.055
Log(Stable Configs. + 1)	0.619	0.056	0.358	0.310
MDB	0.384	0.274	0.195	0.590
Total Solutions	-0.194	0.592	-0.364	0.301
Large Pieces	0.011	0.975	0.078	0.831
Unique Shapes	0.525	0.120	0.495	0.145
IPE	0.322	0.365	0.188	0.603

Overall, these results suggest that, despite the small sample size, human difficulty perception in this domain can be approximated through structural heuristics, ensuring the game remains appropriately challenging for players.

6. Conclusion

This paper presents an analysis of the human perception of level difficulty using a qualitative solver for the Camelot Jr. bridge-building puzzle game. For this, we employ a three-way triangulation strategy to examine the same metrics using third-party expert-labeled data, subjective user ratings, and objective time spent on each level. Our results reveal consistent evidence for various difficulty metrics across the initial grid and dynamic factors.

Even though there are sources of error across all the approaches we tried, we attempted to mitigate them by randomizing the order in which problems are shown to players and by using pruning strategies to eliminate spurious player ratings. More importantly, the similarity of trends across the three strategies strengthens the overall findings. Specifically, *Inventory Size*, *Height Sum*, *Total Volume* and *Stable Configs.* stand out among other factors, with *Unique shapes* also close behind. Some of these features show correlation with each other, even though their sources of origin are different properties.

Solution grids are difficult to analyze in this game because the rules require all inventory blocks to be used for solutions, reducing the scope for creativity in solution choices.

The findings are especially relevant for the spatial reasoning community and for children's pedagogy in general, where training spatial skills early on has been associated with later success in STEM fields. Camelot Jr. involves spatial visualization, spatial path planning, and support affordances, which can be further studied using cognitive tests to examine the effects of playing on player skills.

As future work, we intend to explore our metrics within a mixed reality framework, where spatial bridge-building puzzles could be presented to the user to understand how virtualization affects human performance (e.g., via Meta Quest 3). The difficulty metrics can be extended to explore procedural generation of game puzzles to make them more interesting, offering more diverse new problems that appropriately challenge the player. We also envision the usability of our Camelot Jr. qualitative solving mechanism to enable a hint system for the players, which can advise them on *stability* and *support* schemas, guiding them like a human tutor using natural language or giving visual feedback by highlighting useful blocks and right locations.

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